

Atomic-Scale Assembly of Quantum Defects in Two-Dimensional Semiconductors with the Scanning Tunneling Microscope

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Point defects in two-dimensional (2D) semiconductors hold great potential for quantum technologies, such as spin-qubits or single-photon emitters [1–3] and many candidates have been theoretically identified in the past. However, controlled experimental realization and characterization remains challenging, as it requires creation of well-defined structures embedded in an otherwise pristine material.

Here we present how the scanning tunneling microscope (STM) can be employed to fabricate, characterize and manipulate single point defects and defect complexes on the atomic scale in the 2D semiconductors MoS₂ and WS₂. With the help of transition metal adatoms that are evaporated onto the 2D materials at low temperature [4], we create basal plane substitutional defects M_S, in which a single sulfur atom is replaced by the transition metal. A special focus lies on the Mo_S antisite defect in MoS₂, which is predicted to host a spin triplet ground state that could serve as a spin qubit [1, 2]. In WS₂ we create sulfur vacancies that are assembled into vacancy complexes by making use of a sulfur atom diffusion process induced by inelastic tunneling. Some of these complexes are predicted to be bright color centers, depending on the underlying symmetry [3]. Scanning tunneling spectroscopy is employed to characterize the emergent hybrid in-gap states of the various complexes. Density functional theory calculations are instrumental in interpretation of the experimental results and provide insight into the electronic structure of the defects. Emergent opportunities in the field of single spin control of atomic defects in 2D semiconductors created by STM are demonstrated in recent experiments [5].

[1] Tsai et al., *Nat. Commun.* **13**, 492 (2022).

[2] Lee et al., *Nat. Commun.* **13**, 7501 (2022).

[3] Sun et al., *Nat. Commun.* **15**, 9476 (2024).

[4] Jansen et al., *Phys. Rev. B* **109**, 195430 (2024).

[5] Au-Yeung et al., *arXiv:2602.22301* submitted (2026).